

provides better productivity to data scientists in terms of full flexibility in designing custom analytics and data independence from input formats and physical data representations.

- **Multiple execution modes:** SystemML can be operated in standalone mode on a single machine, allowing data scientists to develop algorithms locally without the need of a distributed cluster. It features a Spark MLContext API that allows for programmatic interaction via Scala, Python, and Java. SystemML also features an embedded API for scoring models.
- **Support of a great many machine learning algorithms:** SystemML features a suite of production-level examples that can be grouped into six broad categories — descriptive statistics, classification, clustering, regression, matrix factorisation and survival analysis.
- **Optimisation:** Algorithms specified in DML are dynamically compiled and optimised based on data and cluster characteristics using rule-based and cost-based optimisation techniques.

Official website: <https://systemml.apache.org/>

Latest version: 1.1.0 for Spark 2.0.2

3. H2O

H2O was designed and developed by *H2O.ai* in 2011. It enables end users to test thousands of prediction models to discover varied patterns in data sets. H2O uses the same types of interfaces like R, Python, Scala, JAVA, JSON and Flow notebook/Web interface, and works in a seamless manner with a large number of Big Data technologies like Hadoop and Spark. It provides a GUI driven platform to companies to perform faster data computations. Currently, H2O supports all types of basic and advanced algorithms like

deep learning, boosting, bagging, naïve Bayes, principal component analysis, time series, k-means, generalised linear models, etc.

Recently, APIs for R, Python, Spark and Hadoop have also been released by H2O, which provide data structures and methods suitable for Big Data. H2O allows users to analyse and visualise whole sets of data without using the Procrustean strategy of studying only a small subset with a conventional statistical package.

H2O uses iterative methods that provide quick answers using all of the client's data. When a client cannot wait for an optimal solution, it can interrupt the computations and use an approximate solution. In its approach to deep learning, H2O divides all the data into subsets and then analyses each subset simultaneously using the same method. These processes are combined to estimate parameters by using the Hogwild scheme, a parallel stochastic gradient method. These methods allow H2O to provide answers that use all the client's data, rather than throwing away most of it and analysing a subset with conventional software.

Features:

- **Very fast and accurate:** H2O is enabled to run fast serialisation between nodes and clusters. It supports large data sets and is highly responsive.
- **Scalable:** Fine-Grain distributed processing on big data at speeds up to 100x faster is done with fine-grain parallelism, which enables optimal efficiency, without introducing degradation in computational accuracy.
- **Easy to use.**
- **Multi-platform:** Can run on Windows, Linux and even Mac OS X.
- **Excellent GUI:** The GUI is compatible with all types of browsers like Safari, Firefox, Internet Explorer and Chrome.

Official website: <https://www.h2o.ai>

Latest version: 3.10.2.2 (Tutte)

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